

Complete JDBC Setup Guide

What is JDBC?

JDBC (Java Database Connectivity) is an API that allows Java programs to connect with databases like:

- MySQL
- PostgreSQL
- Oracle
- SQL Server

In this guide, we'll use **MySQL** because it's the most commonly used database for students and interviews.

Software Required

Software	Purpose	Download Link
JDK (Java Development Kit)	Run Java Programs	https://www.oracle.com/java/technologies/downloads/
IntelliJ IDEA Community Edition	Java IDE	https://www.jetbrains.com/idea/download
MySQL Community Server	Database	https://dev.mysql.com/downloads/mysql/
MySQL Workbench	Database GUI Tool	https://dev.mysql.com/downloads/workbench/
MySQL JDBC Driver (Connector/J)	Connect Java with MySQL	https://dev.mysql.com/downloads/connector/j/

Step 1: Install Java (JDK)

Check if Java is Already Installed

Open Command Prompt:

```
java -version
```

If Java is installed, you'll see something like:

```
java version "21"
```

If Java is Not Installed

Download:

<https://www.oracle.com/java/technologies/downloads/>

Install JDK 21 (or latest LTS version).

Verify Installation

Open Command Prompt:

```
java -version
```

```
javac -version
```

Expected Output:

```
java 21
```

```
javac 21
```

Step 2: Install IntelliJ IDEA

Download:

<https://www.jetbrains.com/idea/download>

Choose:

IntelliJ IDEA Community Edition

Install with default settings.

Launch IntelliJ after installation.

Step 3: Install MySQL Server

Download:

<https://dev.mysql.com/downloads/mysql/>

Choose:

MySQL Community Server

Install it.

During Installation

Select:

Developer Default

or

Server Only

Both work.

Set Root Password

You'll be asked to create:

root password

Example:

root123

Remember this password carefully.

You will need it in JDBC code.

Finish Installation

Keep clicking:

Next → Execute → Finish

Step 4: Install MySQL Workbench

Download:

<https://dev.mysql.com/downloads/workbench/>

Install normally.

Test Database Connection

Open MySQL Workbench.

Login using:

Username: root

Password: root123

If connected successfully, MySQL is working.

Step 5: Create Database

Open MySQL Workbench.

Click:

New Query Tab

Execute:

```
CREATE DATABASE studentdb;
```

Verify:

```
SHOW DATABASES;
```

You should see:

```
studentdb
```

Step 6: Create a Table

Run:

```
USE studentdb;
```

```
CREATE TABLE students(  
  id INT PRIMARY KEY,  
  name VARCHAR(50),  
  city VARCHAR(50)  
);
```

Verify:

```
SHOW TABLES;
```

Output:

```
students
```

Step 7: Download JDBC Driver

Download:

<https://dev.mysql.com/downloads/connector/j/>

Choose:

Platform Independent

Download ZIP file.

Extract ZIP

Example location:

D:\JDBC Driver\

Inside you'll find:

mysql-connector-j-9.x.x.jar

This JAR file is important.

Step 8: Create Java Project in IntelliJ

Open IntelliJ.

Click:

New Project

Select:

Java

Choose installed JDK.

Click:

Next → Finish

Project created successfully.

Step 9: Add JDBC Driver to IntelliJ

In IntelliJ:

File



Project Structure



Libraries



+



Java

Browse to:

mysql-connector-j-9.x.x.jar

Select it.

Click:

OK

Apply

OK

Driver added successfully.

Step 10: Create Main Class

Create:

```
public class Main {  
    public static void main(String[] args) {  
  
    }  
}
```

Step 11: Write JDBC Connection Code

```
import java.sql.Connection;  
import java.sql.DriverManager;
```

```
public class Main {
```

```
public static void main(String[] args) {  
  
    String url = "jdbc:mysql://localhost:3306/studentdb";  
    String username = "root";  
    String password = "root123";  
  
    try {  
  
        Connection con =  
            DriverManager.getConnection(  
                url,  
                username,  
                password);  
  
        System.out.println("Connected Successfully");  
  
        con.close();  
  
    } catch (Exception e) {  
        System.out.println(e);  
    }  
}  
}
```

Step 12: Run Program

Click:

Run Button ►

Expected Output:

Connected Successfully

Congratulations!

Your JDBC connection is working.

JDBC URL Breakdown

`jdbc:mysql://localhost:3306/studentdb`

Part	Meaning
<code>jdbc</code>	JDBC protocol
<code>mysql</code>	Database type
<code>localhost</code>	Your computer
<code>3306</code>	MySQL default port
<code>studentdb</code>	Database name

Step 13: Insert Data Using JDBC

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;

public class Main {

    public static void main(String[] args) {

        String url =
            "jdbc:mysql://localhost:3306/studentdb";

        String user = "root";
        String pass = "root123";

        try {

            Connection con =
                DriverManager.getConnection(
                    url,user,pass);

            Statement st =
                con.createStatement();

            String query =
```

```
        "INSERT INTO students VALUES(1,'Avani','Ahmedabad)";

        st.executeUpdate(query);

        System.out.println("Data Inserted");

        con.close();

    } catch (Exception e) {
        System.out.println(e);
    }
}
}
```

Verify Data

Run in MySQL Workbench:

```
SELECT * FROM students;
```

Output:

```
1  Avani  Ahmedabad
```

Common Errors and Solutions

Error 1

No suitable driver found

Solution

JDBC Driver JAR not added properly.

Re-add:

```
mysql-connector-j.jar
```

Error 2

Access denied for user root

Solution

Wrong username/password.

Check:

```
String username="root";  
String password="yourpassword";
```

Error 3

Unknown database

Solution

Database not created.

Run:

```
CREATE DATABASE studentdb;
```

Error 4

Communications link failure

Solution

MySQL Server is not running.

Open:

Services

Start:

MySQL80

Alternative (Recommended) Setup Using Maven

Instead of manually downloading the JDBC JAR, use Maven.

Create a Maven Project in IntelliJ.

Add this inside `pom.xml`:

```
<dependencies>

  <dependency>
    <groupId>com.mysql</groupId>
    <artifactId>mysql-connector-j</artifactId>
    <version>9.3.0</version>
  </dependency>

</dependencies>
```

Click:

Load Maven Changes

IntelliJ automatically downloads the JDBC driver.

This is the method used in most companies and real-world projects.

JDBC Setup Checklist

- ✓ Install JDK
- ✓ Install IntelliJ IDEA
- ✓ Install MySQL Server
- ✓ Install MySQL Workbench
- ✓ Create Database
- ✓ Download JDBC Driver

✓ Add Driver JAR to IntelliJ

✓ Write Connection Code

✓ Run Program

✓ Verify Connection